Name: MEGHA SINGH Class: SOC-8 Batch: B

Subject: IDS Roll No.: 32 Exp. No.:

Name of the Experiment: ASSIGNMENT - 2

Marks	Teacher's Signature with date
10	

Performed on:

Submitted on:

Assignment - 2

Q1) Define sequential organization. List characteristics of linear data structures that use sequential organisation.

Ans) Sequential organization refers to the way data elements are stored and accessed in a sequence one after the other.

- Characteristics of linear data structures using sequential organization :-

i) Elements are stored in contiguous memory locations
ii) Elements are accessed in a linear sequence
iii) Each element has a unique index or position.
iv) Supports operations like traversal, insertion, deletion and searching.

Q2) Explain the concept of an Array as an (ADT) Abstract Data Type. Discuss its components and basic operations.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Ans) An Array is an ADT that represents a collection of elements of the same data type stored in contiguous memory locations.

- Components :-

i) Elements :→ the data items stored in the array
ii) Index :→ the position of each element in the array
iii) Size :→ the total number of elements in the array

- Basic operations :-

i) Insertion :→ adding an element to the array
ii) Deletion :→ removing an element from the array
iii) Searching :→ finding an element in the array
iv) Traversal :→ accessing each element in the array

Q3) Demonstrate how insertion, deletion and searching operations are performed on a one dimentional array with suitable examples.

Ans) array ⇒ arr = [1, 2, 3, 4, 5]

- Insertion : Insert 6 at index 2
  ↳ Shift elements to the right : arr = [1, 2, 6, 3, 4, 5]

- Deletion : delete element at index 2
  ↳ Shift elements to the left : arr = [1, 2, 3, 4, 5]

- Searching : find element 4
  ↳ Traverse the array : found at index 2

Q4) Analyse the advantages and limitations of using arrays for data storage. Compare array operations such as insertion, deletion and searching based on their efficiency

Ans) • Advantages:-

i) Fast access and modification of elements
ii) Memory efficient
iii) ~~Memory~~ Cache friendly

• Limitations :-

i) Fixed size
ii) Insertion and deletion can be expensive

• Efficiency comparison :-

Operation	Time Complexity
Insertion	O(n)
Deletion	O(n)
Searching	O(n)

Q5) Design a two dimentional array structure to store and process student academic data. Explain how the data can be accessed and manipulated efficiently. Extend your discussion to the use of n-dimentional arrays in real world applications.

Ans) student_data = [[90, 85, 95] # student 1
[80, 90, 85] # student 2
[95, 90, 80] # student 3
]

- Accessing data : student_data [1][2] accesses the marks of student 2 in subject 3

- Manipulating data : update marks using student_data [1][2] = 92.

- n-dimentional arrays are used in :—
  - ↳ Image processing :→ 3D arrays for RGB images
  - ↳ Machine learning :→ high dimentional arrays for feature representation.